

CLASSICAL KUIPER BELT OBJECTS

The objects of the Classical Kuiper Belt, often called KBOs, form several distinct groups that have different compositions and probably originate from different parts of the solar system. One distinction is between “cold” and “hot” KBOs. Despite their name, these groups are identified not by differences in their surface temperatures but by the shape and tilt of their orbits. Cold KBOs have relatively circular orbits with shallow tilts. They also have reddish surfaces, indicating the presence of methane ice. Hot KBOs, such as Makemake, follow more eccentric and tilted orbits and have bluish-white surfaces. Cold KBOs are thought to have originated in roughly the same region where they currently orbit, while hot KBOs probably originated closer to the Sun than they are now. A third group, known as Plutinos, occupy stable orbits in a 2:3 resonance with Neptune. (That is, they orbit the Sun twice for every three orbits of Neptune.) This configuration protects them from Neptune’s gravitational influence and ensures their orbits remain stable. However, the Plutinos, which include Haumea and Pluto itself, are not considered to be Classical KBOs by some astronomers.



MAKEMAKE

Discovered in 2005, Makemake has an estimated diameter of 845–920 miles (1,360–1,480 km), about two-thirds the size of Pluto. With a temperature of only about -405°F (-243°C), Makemake’s surface is covered with methane, ethane, and possibly nitrogen ices.

HAUMEA

With a long axis of about 1,218 miles (1,960 km) and a short axis only half this length, Haumea is unusually elongated for a KBO. It also has a very short rotational period, spinning on its axis once every 4 hours. It was discovered in 2004.



SCATTERED DISK OBJECTS

Beyond the Classical Kuiper Belt is another distinct group of objects, known as Scattered Disk Objects (SDOs). These SDOs move around the Sun in eccentric, often highly tilted orbits that sometimes cross the Classical Belt but also venture much farther out, to 9.3 billion miles (15 billion km) from the Sun or more. They are thought to have originated closer to the Sun and been ejected outward by the gravitational influence of the outer planets. SDOs are still affected by Neptune’s gravity, and the Scattered Disk is thought to be the source of Centaur objects, such as Chiron, as well as some comets. The largest known SDO is Eris, discovered in 2005. Eris has a moon of its own, Dysnomia, whose orbit demonstrates that Eris has about 28 percent more mass than Pluto, despite being probably slightly smaller. Astronomers faced a choice of either promoting Eris to an official 10th planet of the solar system or demoting Pluto, since it was now revealed to be not outstanding within the KBO in terms of either mass or size. They chose the latter option and introduced a new category—dwarf planet—for objects that have planetlike features but lack sufficient gravity to clear their neighboring region of other objects (see p.210).

OORT CLOUD

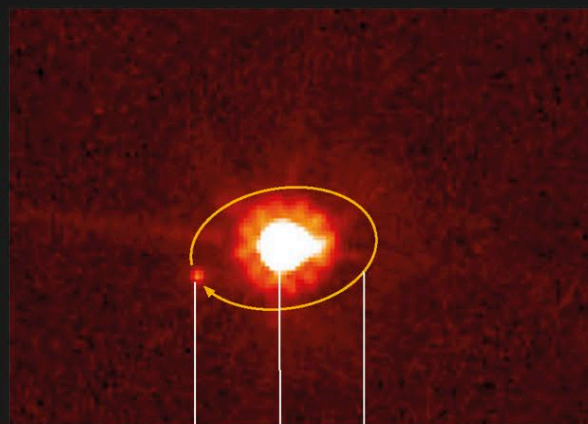
The Oort Cloud is thought to consist of two distinct regions: a spherical, sparsely populated outer cloud and a doughnut-shaped inner cloud. Comets in the more densely populated inner cloud are frequently ejected into the outer cloud and help keep it replenished.

Kuiper Belt

Sun

typical elongated orbit of long-period comet

few comets lie in the region between the inner and outer Oort Cloud



Dysnomia

Eris

orbit of Dysnomia

ERIS AND DYSNOMIA

On September 10, 2005, astronomers using the 32.8-ft (10-m) Keck Telescope in Hawaii discovered that Eris has a moon (seen to the left of Eris), now named Dysnomia. This moon orbits Eris about once every 16 days. Together, Eris and Dysnomia move around the Sun in an eccentric orbit that lasts 557 years.